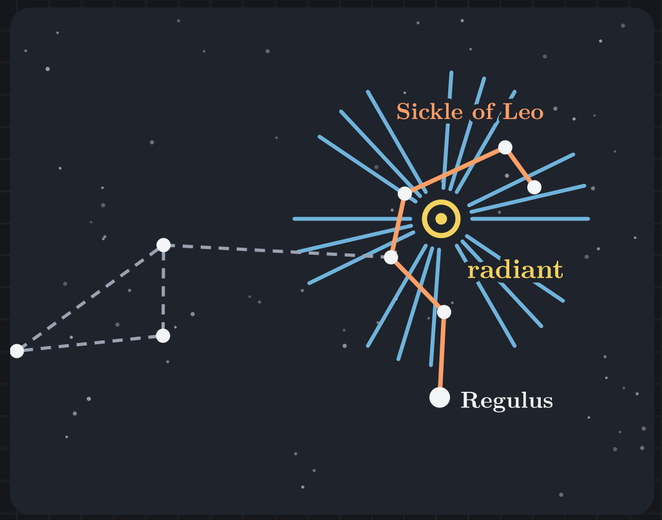
Pi & Planck

SKY WATCH

The Leonid Meteors

Every November, Earth meets dust left by comet Tempel–Tuttle. Some years the sky has stormed.

THE RADIANT



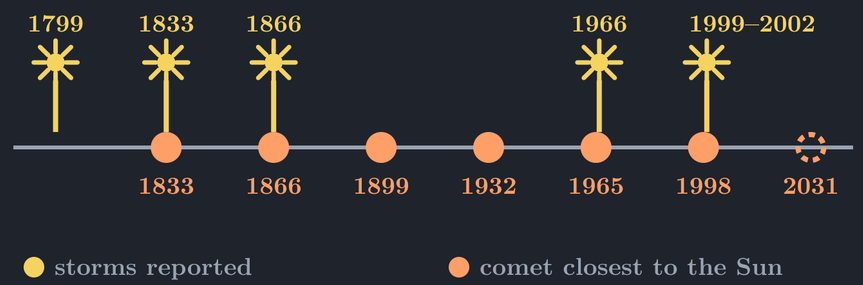
All trails seem to start in the Sickle of Leo: a perspective effect, like parallel rails meeting.

How to watch

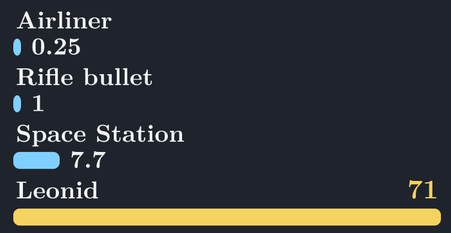
After midnight, away from city lights. No telescope needed.

A COMET ON A 33-YEAR LOOP

Comet 55P/Tempel–Tuttle returns about every 33 years



Speed (km/s)



Tiny grain, big energy

1 mg ⇒ 2.5 kJ

same as a 1 kg ball at 71 m/s

THE SHOW AND THE SCIENCE

Comet dust, not stars

The “shooting star” is a grain, often no bigger than sand, burning up about 100 km above us.

Head-on collision

Earth meets the Leonid stream nearly head-on, so the grains hit at about 71 km/s: among the fastest of the year.

Storms and quiet years

This year the predicted peak is about 15 an hour (ZHR, ideal sky); fewer are usually seen. Some storm years were reported at thousands per hour.

71 km/s typical entry speed, about 255 600 km/h

33 years about one orbit of comet Tempel–Tuttle

15 / hour predicted peak rate (ZHR) this year, under an ideal dark sky

Sources: International Meteor Organization, Meteor Shower Calendar · NASA Meteoroid Environment Office · NASA JPL Small-Body Database, 55P/Tempel–Tuttle · Speeds and energy computed for this graphic